CPE415
Digital Image Processing

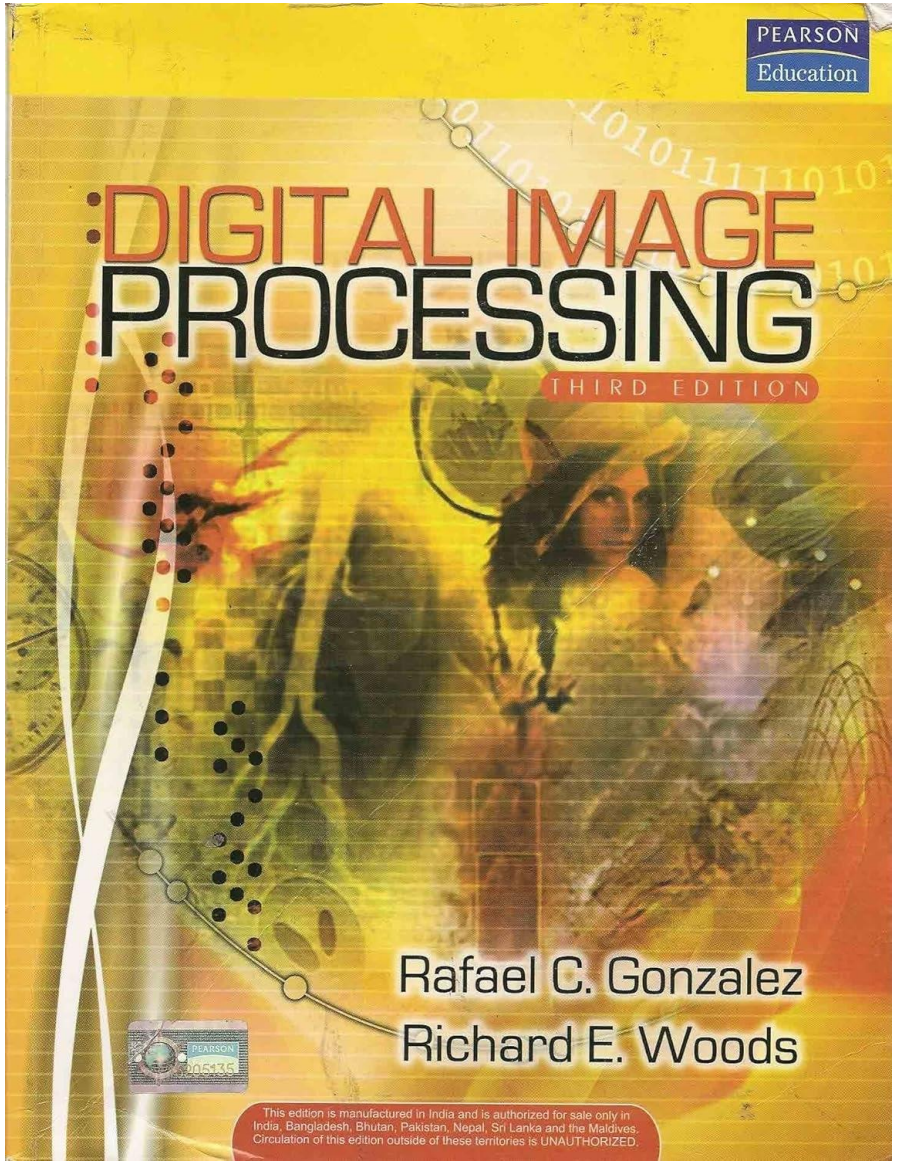
Department of Computer Engineering
COMSATS University Islamabad, Lahore Campus

Text Book

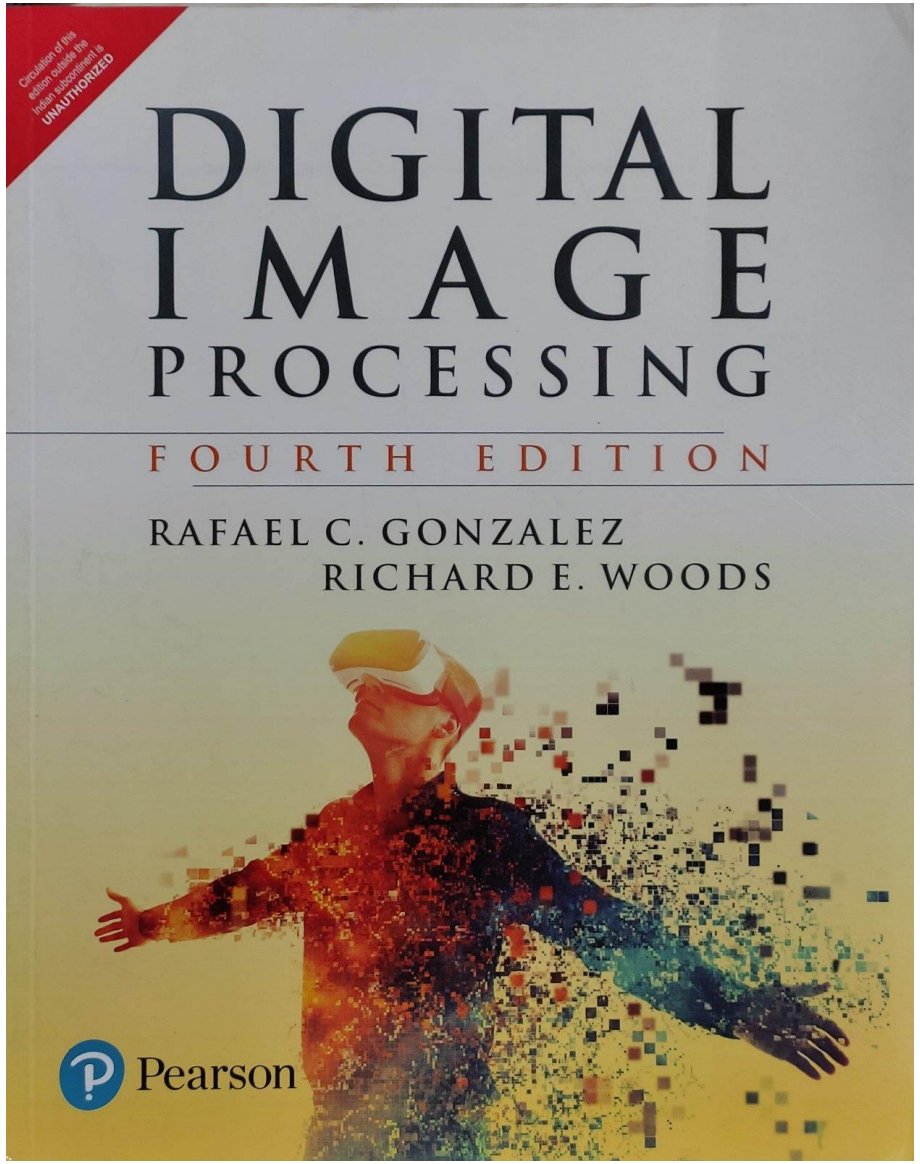
Digital Image Processing

by

Gonzalez & Woods



3rd Edition



4th Edition

Course Outline

- Chapter 1: Introduction
- Chapter 2: Digital Image Fundamentals
- Chapter 3: Image Enhancement in the Spatial Domain
- Chapter 4: Image Enhancement in the Frequency Domain
- Chapter 5: Image Restoration
- Chapter 6: **Color Image Processing**
- Chapter 10: Image Segmentation
- Chapter 11: Representation and Description

>> Few Topics

- Chapter 9: Morphological Image Processing

Attendance & Assessment

Attendance :

- At least **80%** (Both in Theory & Lab) is a must to appear in the terminal exam

  - ✓Emergency case? -> 20% relaxation is already provided for emergency cases!

Assessment :

- 4 Assignments: 10 Marks each (Topics: Chapter End Problems)
- 4 Quizzes: 15 Marks each (Topics: Slides/class discussion/end problems)

- Mid Term Exam: 25 Marks
- Terminal Exam: 50 Marks

Assessment Mapping

Chapter 1: Introduction
Chapter 2: Digital Image Fundamentals
Chapter 3: Image Enhancement in the Spatial Domain
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Chapter 6: Color Image Processing
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Chapter 10: Image Segmentation
Chapter 11: Representation and Description

Q1,Q2
A1,A2
Midterm

Q3,Q4
A3,A4
Terminal

Terminal